Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The range of the function $f(x) = {}^{7-x}P_{x-3}$ is

(A) $\{1, 2, 3, 4\}$

(B) $\{1, 2, 3, 4, 5, 6\}$

(C) $\{1, 2, 3\}$

(D) $\{1, 2, 3, 4, 5\}$

Q.2

If $f : (0, \pi) \to \mathbf{R}$ is given by $f(x) = \sum_{k=1}^{n} [1 + \sin kx]$, $[x]$ denotes the greatest integer function, then the range of $f(x)$ is

(A) $\{n - 1, n + 1\}$

(B) $\{n\}$

(C) $\{n, n + 1\}$

(D) $\{n - 1, n\}$

Q.3

Let $f(x) = x^4 - 2x^2 + 5$ be defined on $[-2, 2]$
Statement-1: The range of $f(x)$ is $[2, 13]$ Statement-2: The greatest value of f is attained at $x = 2$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.4

The value of a for which one root of the equation $x^2 - (a + 1)x + a^2 + a - 8 = 0$ exceeds 2 and other is less than 2, are given by

(A) $3 < a < 10$

@IITJEE_Advanced

(B) $a \geq 10$

(C) $-2 < a < 3$

(D) $a \leq -2$

Q.5

Let $f : (1, \infty) \to (1, \infty)$ be defined by $f(x) = \frac{x+2}{x-1}$. Then

(A) f is one-one and onto

(B) f is one-one but not onto

(C) f is not one-one but onto

(D) f is neither one-one nor onto

Q.6

Let $f : (-1, 1) \to (0, \pi)$ be defined by $f(x) = \cot^{-1} \frac{2x}{1-x^2}$ Then

(A) f is one-one but not onto

(B) f is onto but not one-one

(C) f is both one-one and onto

(D) f is neither one-one nor onto

Q.7

If $(m, \quad n)$ represents the domain of the function defined as $f\left(x\right) = \sqrt{\log_{10} \frac{\log_{10} x}{23 - \log_{10} x}}$. Find $\frac{n}{m}$

(A) 10

(B) 20

(C) 30

(D) 40

Q.8

The range of the function $\mathrm{f(x)} = \mathrm{x}^2 + 2\mathrm{x} + 2$ is

(A) $(1, \infty)$

(B) $(2, \infty)$

(C) $(0, \infty)$

(D) $[1, \infty)$

@IITJEE_Advanced

Q.9

The range of the function $f(x) = \frac{e^x - e^{|x|}}{e^x + e^{|x|}}$ is

(A) $(-\infty, \infty)$

(B) $[0, 1)$

(C) $(-1, 0]$

(D) $(-1, 1)$

Q.10

Let f: R → R and $f_n(x) = ff_{n-1}(x) \quad \forall n \geq 2, \quad n \in N$. Then the roots of the equation $f_3(x) f_2(x) f(x) - 25 f_2(x) f(x) + 175 f(x) = 375$ which also satisfy the equation $f(x) = x$ will be

(A) 5 only

(B) 15 only

(C) 10 only

(D) $5,\ 15$

Q.11

The range of $f(x) = [\sin\{x\}]$ where $\{.\}$ represents the fractional part function, [.] represents greatest the integer function.

(A) $\{0\}$

(B) $\{2\}$

(C) $\{4\}$

(D) None of these

Q.12

Find the range of $f(x) = \text{sgn}\left(x^2 - 2x + 3\right)$ is

(A) $\{-1, 0, 1\}$

(B) $\{0, 1\}$

(C) $\{1\}$

(D) $\{-1, -2, 2, 3\}$

Q.13

@IITJEE_Advanced

The range of $f(x) = \text{sgn}\left(x^2 - 2x + 3\right)$ is

(A) $\{1\}$

(B) $\{2\}$

(C) \{3\}

(D) None of these

Q.14

The range of $f(x) = \frac{x-[x]}{1-[x]+x}$, where [ ] represents the greatest integer fumction.

(A) $\left[0, \frac{1}{2}\right)$

(B) $\left(-\frac{1}{2}, 0\right)$

(C) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

(D) None of these

Q.15

Let $f(x)$ be defined on $[-2, 2]$ and is given by $f(x) = \begin{cases} -1, & -2 \le x \le 0 \\ x - 1, & 0 \le x \le 2 \end{cases}$, then $f(|x|)$ is defined as

(A) $f(|x|) = \begin{cases} 1 & -2 \le x \le 0 \\ 1 - x, & 0 < x \le 2 \end{cases}$

(B) $f(|x|) = x - 1 \forall x \in R$

(C) $f(|x|) = \begin{cases} -x - 1, & -2 \le x \le 0 \\ x - 1, & 0 < x \le 2 \end{cases}$

(D) None of these

Q.16

Number of solutions of the equation $\sin^{-1}\left(\sin 6x\right) = x$ in $x \in [0, \pi]$ is

(A) 4

(B) 5

(C) 6

(D) 7

Q.17

If $\log_{1/2}\left(x^2 - 5x + 7\right) > 0$, then exhaustive range of values of x is

(A) $(-\infty, 2) \cup (3, \infty)$

@IITJEE_Advanced

(B) $(2,3)$

(C) $(-\infty,1)\cup(1,2)\cup(2,\infty)$

(D) None of these

Q.18

The domain of the function
$\sqrt{x^2-5x+6}+\sqrt{2x+8-x^2}$ is

(A) $[2,3]$

(B) $[-2,4]$

(C) $[-2,2]\cup[3,4]$

(D) $[-2,1]\cup[2,4]$

Q.19

Let $f(x)=\frac{\sin x}{\sqrt{1+\tan^2 x}}-\frac{\cos x}{\sqrt{1+\cot^2 x}}$ then range of $f(x)$ is

(A) $[-1,0]$

(B) $[0,1]$

(C) $[-1,1]$

(D) none of these

Q.20

For which real values of x and y, the equation $\sec^2\theta=\frac{4xy}{(x+y)^2}$ is possible ?

(A) $x=y$

(B) $x>y$

(C) $x<y$

(D) None of these

Answers & Solutions

Q.1 Answer:

$\{1,2,3\}$

Solution:

$7-x\geq 1, x-3\geq 0$
and $7-x\geq x-3$
$\Rightarrow x\leq 6, x\geq 3, x\leq 5$
Thus $3\leq x\leq 5$

@IITJEE_Advanced

$\therefore$ **Range** $= \{4P_0, 3P_1, 2P_2\}$
$= \{1, 3, 2\}$

Q.2 Answer:

$\{n, n+1\}$

Solution:

$f(x) = \sum_{k=1}^{n} 1 + [\sin kx] = n + [\sin x] + [\sin 2x] + \cdots$
$+[\sin nx]$. **If** $kx \neq \frac{\pi}{2}$ **for any** $k = 1, 2 \ldots n$ **then 0**
$< kx < \pi$ **and** $kx \neq \pi/2$ **so** $0 < \sin kx < 1$. **Hence**
$[\sin kx] = 0, k = 1, 2 \ldots n$. **i.e.** $f(x) = n$. **If** $kx = \frac{\pi}{2}$
for some k **then** $x = \frac{\pi}{2k}$, **hence** $\sin x, \sin 2x, \ldots, \sin(k-1)x$ **will lie between 0 and 1 so**
$[\sin jx] = 0\ 1 \leq j \leq k-1; \sin kx = 1$ **so** $f(x)$ **can be** $n+1$ **or** n.

Q.3 Answer:

STATEMENT-1 is False, STATEMENT-2 is True

Solution:

$f'(x) = 4x^3 - 4x = 4x(x-1)(x+1)$.
The critical points of f **are** $0, -1, 1$ **But** $f(0) = 5, f(1) = 4, f(-1) =$
$4, f(2) = 13$. **So the range of** f **is** $[4, 13]$ **and grea, test value of** f **is at** $x = 2$

Q.4 Answer:

$-2 < a < 3$

Solution:

Quadrate equation is:
$x^2 - (a+1)x + a^2 + a - 8 = 0$
one root exceeds 2 and other is
lesser than
$\Rightarrow$ **2 lies between the roots.**
A rough diagram it as shown below.
where α **and** β **are**
its roots.
$\Rightarrow$ **For 2 to lie.**
between roots.
The only condition is
$f(2) < 0$
$\Rightarrow 2^2 - (a+1)(2) + a^2 + a - 8 < 0$
$\Rightarrow 4 - 2a - 2 + a^2 + a - 8 < 0$
$\Rightarrow a^2 - a - 6 < 0$
$\Rightarrow a^2 - 3a + 2a - 6 < 0$

@IITJEE_Advanced

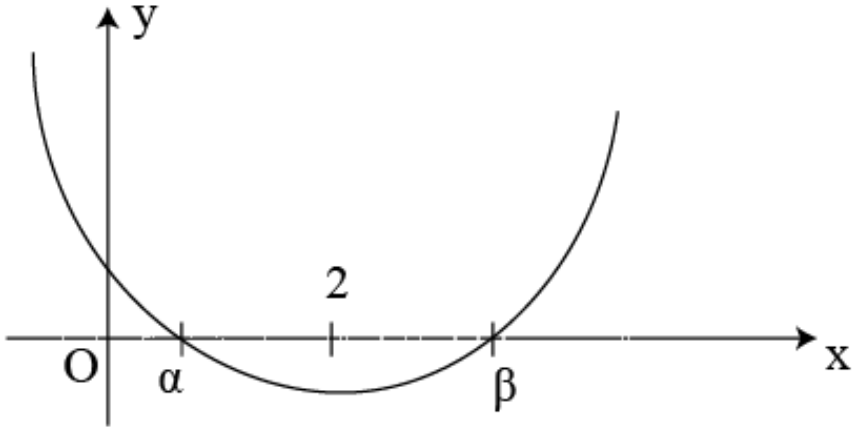
$\Rightarrow a(a-3)+2(a-3)<0$

$\Rightarrow (a+2)(a-3)<0$

$\Rightarrow$ **using sign reheme method.**

ae $(-2,3)$

i.e. $-2<a<3$



Q.5 Answer:

f is one-one and onto

Solution:

If $\frac{x_1+2}{x_1-1}=\frac{x_2+2}{x_2-1}\Rightarrow x_1x_2+2x_2-x_1-2$

$x_1x_2+2x_1-x_2-2$

$\Rightarrow \quad 3x_2=3x_1\Rightarrow x_1=x_2$

So f **is one-one**

If $y=f(x)=\frac{x+2}{x-1}\Rightarrow yx-y=x+2$

$\Rightarrow \quad (y-1)x=y+2$

$\Rightarrow \quad x=\frac{y+2}{y-1}\in(1,\infty)$

Thus f **is onto**

Q.6 Answer:

f is both one-one and onto

Solution:

$f(x)=\cot^{-1}\frac{2x}{1-x^2}$ **is clearly one-one as**

$f(x_1)=f(x_2)$

$\Rightarrow x_1=x_2$

Next, let $y=f(x)\Rightarrow \cot y=\frac{2x}{1-x^2}$

$=\frac{2\tan(\theta/2)}{1-\tan^2(\theta/2)}=\tan\theta$

$\Rightarrow y=\pi/2-\theta=\pi/2-2\tan^{-1}x$. **Taking** $x=\tan(\theta/2)$

so $y\in(0,\pi)\Rightarrow 0<y<\pi$

$\Rightarrow -\pi/4<\tan^{-1}x<\pi/4$

$\Rightarrow -1<x<1\Rightarrow x\in(-1,1)$

and thus f **is onto.**

@IITJEE_Advanced

Q.7 Answer:

10

Solution:

$f(x)$ **is defined if** $\log_{10}\frac{\log_{10}x}{23-\log_{10}x} \geq 0$ **and** $\frac{\log_{10}x}{23-\log_{10}x} > 0$ **and** $x > 0$

$\Rightarrow \frac{\log_{10}x}{23-\log_{10}x} \geq 10^0 = 1$ **and** $\frac{\log_{10}x}{\log_{10}x-3} < 0$ **and** $x > 0$

$\Rightarrow \frac{\log_{10}x-23-\log_{10}x}{23-\log_{10}x} \geq 0$ **and** $\frac{\log_{10}x}{\log_{10}x-3} < 0$

and

$x > 0$

$\Rightarrow \frac{3\log_{10}x-2}{2\log_{10}x-3} \leq 0$ **and** $\frac{\log_{10}x}{\log_{10}x-3} < 0$ **and** $x > 0$

$\Rightarrow 2 \leq \log_{10}x < 3$ **and** $0 < \log_{10}x < 3$ **and** $x > 0$

$\Rightarrow 10^2 \leq x < 10^3$ **i.e.** $100 \leq x < 1000$

$\Rightarrow m = 100, n = 1000$

$\Rightarrow \frac{n}{m} = 10$

Q.8 Answer:

$[1, \infty)$

Solution:

$$f(x) = y = x^2 + 2x + 2 = (x+1)^2 + 1$$

Let $y - 1 = (x+1)^2 \Rightarrow x + 1 = \sqrt{y-1}$

$$x = \sqrt{y-1} - 1$$

Since, $y - 1 \geq 0 \therefore y \geq 1$

$\therefore$ **range is** $[1, \infty)$.

Q.9 Answer:

$(-1, 0]$

Solution:

$$f(x) = \frac{e^x - e^{|x|}}{e^x + e^{|x|}} = \begin{cases} 0, & x \geq 0 \\ \frac{e^x - e^{-x}}{e^x + e^{-x}}, & x < 0 \end{cases}$$

Clearly, $f(x)$ **is identically zero if** $x \geq 0$ **(1)**

If $x < 0$, **let** $y = f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ **or** $e^{2x} = \frac{1+y}{1-y}$

$\because x < 0; e^{2x} < 1$ **or** $0 < e^{2x} < 1$

$\therefore \quad 0 < \frac{1+y}{1-y} < 1$ **or** $\frac{1+y}{1-y} > 0$ **and** $\frac{1+y}{1-y} < 1$

or $(y+1)(y-1) < 0$ **and** $\frac{2y}{1-y} < 0$

i.e., $-1 < y < 1$ **and** $y < 0$ **or** $y > 1$ **or** $-1 < y < 0$ **(2)**

Q.10 Answer:

5, 15

@IITJEE_Advanced

Solution:

$f_2(x) = f(f(x)) = f(x) = x$
$f_3(x) = ff_2(x) = f(x) = x$
$\Rightarrow x^3 - 25x^2 + 175x - 375 = 0$
$(x-5)x^2 - 20x + 75 = 0$
$(x-5)^2(x-15) = 0 \Rightarrow x = 5, 15$

Q.11 Answer:

$\{0\}$

Solution:

$f(x) = [\sin\{x\}]$
Here, $\{x\}$ can take all its possible values and sine function is defined for all real values.
Hence, $0 \le \{x\} < 1$
$\Rightarrow 0 \le \sin\{x\} < \sin 1$
$\Rightarrow [\sin\{x\}] = 0$
Hence, the range is $\{0\}$.

Q.12 Answer:

$\{1\}$

Solution:

We know that $f(x) = Sgn(y) = \begin{cases} 1 & f(x) > 0 \\ 0 & f(x) = 0 \\ -1 & f(x) < 1 \end{cases}$
The given function is $f(x) = \text{sgn}\left(x^2 - 2x + 3\right) = \frac{|x^2-2x+3|}{x^2-2x+3}$
Consider, $x^2 - 2x + 3 = (x^2 - 2x + 1) + 2 = (x-1)^2 + 2 > 0$
So the quadratic equation is always positive
$f(x) = \text{sgn}\left(x^2 - 2x + 3\right) = \frac{|x^2-2x+3|}{x^2-2x+3} = \frac{x^2-2x+3}{x^2-2x+3} = 1$

Q.13 Answer:

$\{1\}$

Solution:

$x^2 - 2x + 3 = (x-1)^2 + 1 > 0 \forall x \in R$
$\Rightarrow f(x) = \text{sign}\left(x^2 - 2x + 3\right) = 1$
Hence, the range is $\{1\}$.

Q.14 Answer:

$\left[0, \frac{1}{2}\right)$

Solution:

@IITJEE_Advanced

$f(x) = \frac{x-[x]}{1-[x]+x} = \frac{\{x\}}{1+\{x\}} = 1 - \frac{1}{1+\{x\}}$

Now, $0 \le \{x\} < 1$

$\Rightarrow 1 \le \{x\} + 1 < 2$

$\Rightarrow \frac{1}{2} < \frac{1}{1+\{x\}} \le 1$

$\Rightarrow -1 \le -\frac{1}{1+\{x\}} < -\frac{1}{2}$

$\Rightarrow 0 \le 1 - \frac{1}{1+\{x\}} < \frac{1}{2}$

$$0 \leqslant \frac{\{x\}}{1+\{x\}} < \frac{1}{2}$$

$$0 \leqslant f(x) < \frac{1}{2}$$

Q.15 Answer:

$$f(|x|) = \begin{cases} -x-1, & -2 \le x \le 0 \\ x-1, & 0 < x \le 2 \end{cases}$$

Solution:

We have $f(x) = \begin{cases} -1, -2 \le x \le 0 \\ x-1, 0 \le x \le 2 \end{cases}$

$f(|x|) = \begin{cases} -1, -2 \le |x| \le 0 \\ |x|-1, 0 \le |x| \le 2 \end{cases} \Rightarrow f(|x|) = |x|-1, 0 \le |x| \le 2$

(as $-2 \le |x| \le 0$ **is not possible)** $\Rightarrow f(|x|) = \begin{cases} -x-1, & -2 \le x \le 0 \\ x-1, & 0 < x \le 2 \end{cases}$

Q.16 Answer:

4

Solution:

$\sin^{-1}\sin 6x = x$

$\Rightarrow \sin 6x = \sin x$

$\Rightarrow x = \frac{\pi}{7}, \quad \frac{3\pi}{7}, \quad \frac{5\pi}{7}, \quad \pi, \quad 0, \quad \frac{2\pi}{5}, \quad \frac{4\pi}{5}$

But $\frac{5\pi}{7}, \pi$ **and** $\frac{4\pi}{5}$ **are rejected as any value of** x

which is the solution of a given equation can't exceed

$\frac{\pi}{2}$.

Q.17 Answer:

$(2, 3)$

Solution:

$f(x) = \log_{1/2}\left(x^2 - 5x + 7\right) > 0 \Rightarrow x^2 - 5x + 7 > 0$

$x^2 - 5x + 7 < 1, x \in \mathrm{R} \Rightarrow x^2 - 5x + 6 < 0 \Rightarrow x \in (2, 3)$

Q.18 Answer:

@IITJEE_Advanced

$[-2,2] \cup [3,4]$

Solution:

$f(x) = \sqrt{(x-2)(x-3)} + \sqrt{-(x-4)(x+2)}$

The first part is real outside $(2,3)$ **and the second is real in** $[-2,4]$ **so that the domain is** $[-2,2] \cup [3,4]$.

Q.19 Answer:

$[-1,1]$

Solution:

$f(x) = \frac{\sin x}{|\sec x|} - \frac{\cos x}{|\cos ecx|}$

Range $f(x) = \sin x \cdot |\cos x| - \cos x |\sin x| = ?$

$$f(x) = \begin{cases} 0 & x \in \left[0, \frac{\pi}{2}\right] \\ -\sin 2x & x \in \left(\frac{\pi}{2}, \pi\right) \\ 0 & x \in \left(\pi, \frac{3\pi}{2}\right) \\ \sin 2x & x \in \left(\frac{3\pi}{2}, 2\pi\right) \end{cases}$$

$-\sin 2x$	o
o	$\sin 2x$

So range is $[-1,1]$

Q.20 Answer:

$x = y$

Solution:

Given equation, $\sec^2\theta = \frac{4xy}{(x+y)^2}$

Since range of $\sec\theta$ **is** $(-\infty,-1] \cup [1,\infty)$.

$\sec^2\theta \geq 1 \quad \Rightarrow \quad \frac{4xy}{(x+y)^2} \geq 1$

$\Rightarrow \quad (x+y)^2 - 4xy \leq 0 \Rightarrow x^2 + y^2 + 2xy - 4xy \leq 0$

$\Rightarrow \quad (x-y)^2 \leq 0$ **But** $(x-y)^2 < 0$ **for any** $x, y \in \mathrm{R}$

$\therefore \quad (x-y)^2 = 0 \Rightarrow x = y$

@IITJEE_Advanced